

The Observation Limit, and the Three Layers of the Null

An Attack on the Resolution-Poset Compression, and What Survives

Matěj Rada

MatejRada@email.cz

Licensed under CC BY-NC-ND 4.0

September 12, 2026

Abstract

A proposed compression of the ignorance framework would make “resolution” a single directed poset $\rho = (S, \pi, M)$ of sensor, precision, and representation, form the total map $\mathbf{F} = \prod_{\rho} F_{\rho}$, and define ignorance as the persistent kernel $\mathcal{I}_x = \bigcap_{\rho} \ker dF_{\rho} = \ker d\mathbf{F}$. We put this to the falsification test. Three attacks succeed. **Precision** is not a map-layer refinement: the true differential dF is precision-independent, so a below-threshold direction has $dF v \neq 0$ exactly and is *not* in $\ker dF$; it lives in the gap between the true and the computed differential. **Representation** is not a map-layer refinement either: for $F = (x, x^2)$ the information is present in $\text{Im } F$ (full rank), and the “null” appears only in a downstream construction A (a linear fit), i.e. in $\ker A$, not $\ker dF$. **Directedness** fails under complementarity: non-commuting measurements have no common refinement, so the product map is not operationally realizable and $\mathbf{F}$ presumes a joint measurability that quantum mechanics denies. Hence the single-poset unification conflates three logically distinct layers, and—as the proposal itself concedes—introduces no new primitive. What survives is a clean, established object: the *observation limit* F_{∞} , the inverse limit of a *compatible* (jointly-refinable) sensor-refinement system, with $\mathcal{I}_x = \ker dF_{\infty}$ the sensor-persistent null. It is well-defined, attained at finite resolution in finite dimensions, and invariant under every downstream construction and every precision—the true physical wall. The correct picture is three layers, not one poset: the map (sensors) carries the wall; evaluation (precision, sampling) and downstream construction (representation) carry only removable nulls, removable precisely because their information already lies in $\text{Im } F_{\infty}$.

Contents

1	The proposed compression	1
2	Three attacks	2
3	What survives: the observation limit	2
4	The corrected architecture: three layers, one wall	2
5	Non-claims	3
6	Conclusion	3

1 The proposed compression

The proposal is attractive and partly correct. It makes resolution a directed family $\{F_\rho : X \rightarrow Y_\rho\}$ with coarse-graining compatibility $F_\rho = q_{\rho'\rho} \circ F_{\rho'}$, defines the persistent ignorance space

$$\mathcal{I}_x = \bigcap_{\rho} \ker dF_\rho(x) = \ker d\mathbf{F}_x, \quad \mathbf{F} = \prod_{\rho} F_\rho,$$

and recovers Fisher as a derived detector ($\text{rad } g_\rho^F = \ker dF_\rho$ under regularity). Two of these moves are right and we keep them: making the sensor family directed with a coarse-graining axiom, and treating Fisher as measuring, not defining, distinguishability. The error is the *scope* of the intersection.

2 Three attacks

Attack 1 (Precision is not a map-layer refinement). The differential dF is exact and independent of the arithmetic used to evaluate it. Take $dF = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 10^{-10} & 0 \end{pmatrix}$ and $v = e_2$: then $dF v = 10^{-10} e_2 \neq 0$ *exactly*, so $v \notin \ker dF$. Finite-precision evaluation of $g^F = dF^\top \Sigma^{-1} dF$ returns an eigenvalue $\sim 10^{-20}$ and rounds v into a *computed* kernel, but the true $\ker dF$ —and hence $\bigcap_{\rho} \ker dF_\rho$ over precision—is unchanged. Precision therefore contributes nothing to $\mathcal{I}_x$; it lives in the gap between the true and the computed differential.

Attack 2 (Representation is not a map-layer refinement). Let $F = (x, y)$ with $y = x^2 + \text{noise}$. The observed columns are full rank: the information is in $\text{Im } F$, and $\ker dF$ has no contribution from the quadratic structure. The “null” appears only when a downstream construction A —a linear fit $y \sim x$ —is applied ($R^2 \approx 0$). Thus representation-null is $\ker A$ on $\text{Im } F$, not $\ker dF$; it is a property of the analyst’s map A , not of the observation map.

Attack 3 (Directedness fails under complementarity). $\mathbf{F} = \prod_{\rho} F_\rho$ presumes that any two resolutions have a common refinement (a performable joint measurement). For non-commuting observables this fails: $[\sigma_z, \sigma_x] \neq 0$ means no simultaneous eigenbasis and no realizable product map $F_z \times F_x$. The directed-family hypothesis, and with it the existence of a single limit map over *all* resolutions, is not available in general; it requires a *compatible* sub-family.

Corollary 1 (The single poset conflates three layers). *Sensor refinement acts on the observation map F ; precision acts on the evaluation of F ; representation acts on a downstream construction A on $\text{Im } F$. Only the sensor axis lives in $\ker dF$. The union of the three into one poset $\rho = (S, \pi, M)$ with a single intersection $\bigcap \ker dF_\rho$ is therefore ill-typed, and no new primitive is introduced—the surviving object is a kernel of a limit of linear maps.*

3 What survives: the observation limit

Definition 1 (Compatible sensor system and observation limit). A sensor-refinement system is a directed set of resolutions with smooth coarse-graining $F_S = q_{S'S} \circ F_{S'}$ for $S \preceq S'$, all jointly performable (*compatible*). Its observation limit is the inverse limit $F_\infty = \varprojlim_S F_S : X \rightarrow \varprojlim_S Y_S$, and the persistent (sensor) null is $\mathcal{I}_x = \ker dF_\infty(x) = \bigcap_S \ker dF_S(x)$.

Proposition 1 (Well-posedness and finite attainment). *Over a compatible system in finite dimensions, $\mathcal{I}_x = \ker dF_{S^*}(x)$ for some finite resolution S^* . The residual freedom is admissible gauge.*

Proof. $\dim \ker dF_S$ is a non-negative integer, non-increasing under refinement by the coarse-graining axiom (which gives $\ker dF_{S'} \subseteq \ker dF_S$). It attains a minimum at some S^* ; by directedness any S has a common refinement $S'' \succeq S, S^*$ with $\ker dF_{S''} \subseteq \ker dF_{S^*}$ of minimal dimension, hence equal, so the intersection is realized at S^* . $\square$

Remark 1 (No new primitive). F_∞ is an inverse limit and $\mathcal{I}_x$ its kernel—standard category theory and linear algebra. This is the elimination outcome again: the right object exists already; the contribution is the correct identification and the layer separation, not a new symbol.

4 The corrected architecture: three layers, one wall

layer	acts on	null it carries
map (sensors)	the observation map F	$\ker dF_\infty = \mathcal{I}_x$ — the physical wall
evaluation (precision, sampling)	how F is computed/estimated	below-threshold, <i>removable</i>
downstream (representation)	a construction A on $\text{Im } F$	$\ker A$, <i>removable</i>

Theorem 1 (The wall, and the removable layers). *Let $v \in \mathcal{I}_x = \ker dF_\infty(x)$. Then for every downstream construction $A : \text{Im } F_\infty \rightarrow Z$ and every arithmetic precision, $A \circ F_\infty$ is constant along v ; no evaluation or downstream refinement opens $\mathcal{I}_x$. Conversely, a direction with $dF_\infty v \neq 0$ (its information lies in $\text{Im } F_\infty$) is opened by sufficient precision, sufficient data, or a richer A . Thus the physical wall is exactly the map-layer kernel; the other two layers carry only removable nulls.*

Proof. $v \in \ker dF_\infty$ makes F_∞ constant along v , so $A \circ F_\infty$ is constant along v by the chain rule; precision changes the evaluation of $A \circ F_\infty$, not its constancy. If instead $dF_\infty v \neq 0$, the value F_∞ varies along v , so a precise enough evaluation, enough samples, or an A retaining that variation recovers it. $\square$

Remark 2 (Complementarity is a real boundary). F_∞ exists over a compatible family. When the admissible resolutions include mutually incompatible measurements, there is no total limit and $\mathcal{I}_x$ must be taken relative to a chosen compatible sub-family. This is not a defect to be patched away: it is where quantum measurement enters the ignorance question, and it should be stated as a hypothesis, not assumed silently.

5 Non-claims

Non-claim 1 (No new primitive). The observation limit is an inverse-limit kernel. Nothing here is a new mathematical object; the proposal’s own instinct against a “resolution stack” primitive is correct.

Non-claim 2 (Not one poset). Sensor, precision, and representation are three layers, not one resolution order. Only the sensor layer contributes to $\ker dF_\infty$.

Non-claim 3 (Compatibility required). The total observation limit presumes a jointly-refinable family. Under complementarity it does not exist; $\mathcal{I}_x$ is then relative to a compatible sub-family.

Non-claim 4 (Finite dimensions for finite attainment). Finite attainment of the persistent kernel uses finite-dimensional tangent spaces; infinite-dimensional latent spaces may not stabilize at finite resolution.

6 Conclusion

The compression was worth attacking, and the attack pays. Its correct kernel is the observation limit F_∞ over a compatible sensor family, with $\mathcal{I}_x = \ker dF_\infty$ the sensor-persistent null—an established object, no new primitive, attained finitely, and invariant under every precision and

every downstream construction: the physical wall. Its error is folding precision and representation into the same intersection; those are separate layers (evaluation, and downstream A) whose nulls are removable exactly because their information already lies in $\text{Im } F_\infty$. And its silent assumption is joint measurability, which complementarity denies. The defensible foundation is therefore smaller than proposed and sharper: one map-layer wall, two removable layers, and one honest hypothesis (compatibility) marking where the total limit ceases to exist.

References

- [1] M. Rada, *The Law and Boundary of Observational Ignorance*, 2026.
- [2] M. Rada, *Resolution Geometry: Reconstruction and Identifiability*, 2026.
- [3] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer, 1998 (limits).
- [4] S. Amari and H. Nagaoka, *Methods of Information Geometry*, AMS/Oxford, 2000.
- [5] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, 2000 (complementarity).